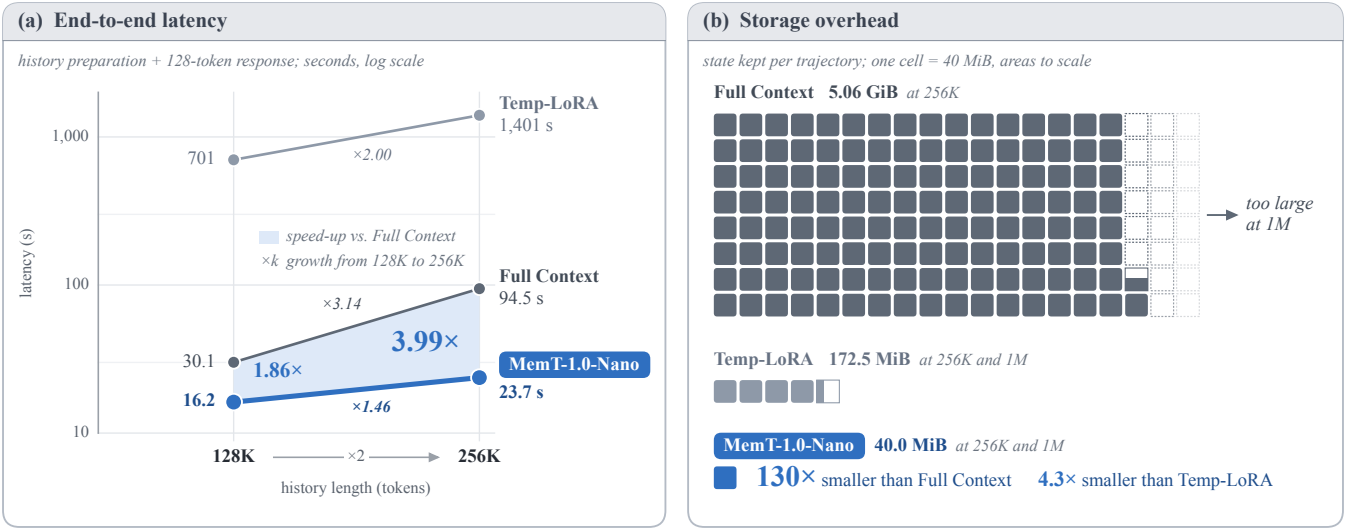

(a) End-to-end latency
history preparation + 128-token response; seconds, log scale
Temp-LoRA
1,401 s
1,000
701
×2.00
speed-up vs. Full Context
×k growth from 128K to 256K
Full Context
94.5 s
×3.14
3.99×
30.1
1.86×
MemT-1.0-Nano
16.2
23.7 s
×1.46
100
10
latency (s)
128K
×2
256K
history length (tokens)
(b) Storage overhead
state kept per trajectory; one cell = 40 MiB, areas to scale
Full Context 5.06 GiB at 256K
too large
at 1M
Temp-LoRA 172.5 MiB at 256K and 1M
MemT-1.0-Nano 40.0 MiB at 256K and 1M
130× smaller than Full Context
4.3× smaller than Temp-LoRA